THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 3

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A point P is 10 cm from the centre O of a circle of radius 6 cm. The length of the tangent drawn from P to the circle is: 1

(a) 4 cm **(b)** 8 cm **(c)** 64 cm **(d)** 16 cm

2. A quadrant of a circle has radius 7 cm. The perimeter of the quadrant is: 1

(a) 18 cm **(b)** 11 cm **(c)** 36 cm **(d)** 25 cm

3. The value of $\cos 60^\circ$ is: 1

(a) 2 **(b)** 1 **(c)** $\frac{1}{2}$ **(d)** 0

4. From a point on level ground, 8 m away from the foot of a vertical tower, the angle of elevation of its top is 60°. The height of the tower is: 1

(a) $8\sqrt{3}$ m **(b)** $\frac{8\sqrt{3}}{3}$ m **(c)** $16\sqrt{3}$ m **(d)** 8 m

5. The length of the tangent from an external point P to a circle is 12 cm and P is 13 cm from the centre. The radius of the circle is: 1

(a) 13 cm **(b)** 1 cm **(c)** 5 cm **(d)** 25 cm

6. The radius of a circle is 14 cm. The area of the circle is: 1

(a) 88 cm² **(b)** 44 cm² **(c)** 1232 cm² **(d)** 616 cm²

7. The value of $\sin^2 35^\circ + \cos^2 35^\circ$ is: 1

(a) 2 **(b)** 35 **(c)** 1 **(d)** 0

8. From a point on level ground, 12 m away from the foot of a vertical tower, the angle of elevation of its top is 30°. The height of the tower is: 1

(a) 12 m **(b)** $4\sqrt{3}$ m **(c)** 4 m **(d)** $12\sqrt{3}$ m

9. A tangent to a circle of radius 15 cm has length 20 cm from the external point P to the point of contact. The distance of P from the centre of the circle is: 1

(a) 5 cm **(b)** 50 cm **(c)** 35 cm **(d)** 25 cm

10. The radius of a circle is 14 cm. The circumference of the circle is: 1

(a) 616 cm **(b)** 176 cm **(c)** 44 cm **(d)** 88 cm

11. $\sin 40^\circ = \cos$ of: 1

(a) 55° **(b)** 50° **(c)** 40° **(d)** 140°

12. A vertical pole is 15 m high. The angle of elevation of the sun is 45° and the pole casts a shadow on level ground. The length of the shadow is: 1

(a) 15 m **(b)** 30 m **(c)** $15\sqrt{3}$ m **(d)** $\frac{15}{2}$ m

13. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 6 cm. The length of each tangent is 8 cm. The perimeter of the quadrilateral $PAOB$ is: 1

(a) 28 cm **(b)** 14 cm **(c)** 32 cm **(d)** 16 cm

14. The circumference of a circle is 88 cm. The radius of the circle is: 1

(a) 44 cm **(b)** 7 cm **(c)** 14 cm **(d)** 28 cm

15. If $\sin A = \frac{3}{5}$, then $\csc A =$ 1

(a) $\frac{3}{5}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{4}{3}$ **(d)** $\frac{5}{3}$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle AOB = 120^\circ$. Find the $\angle APB$. 2

17. A sector of a circle of radius 21 cm has central angle 45°. Find the area of the sector. 2

18. In a right triangle the perpendicular is 12 and the hypotenuse is 13. Find $\sin\theta$. 2

19. From a point on level ground, 16 m away from the foot of a vertical tower, the angle of elevation of its top is 45°. Find the height of the tower. 2

20. A point P is 26 cm from the centre O of a circle of radius 24 cm. Find the length of the tangent drawn from P to the circle. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 7 cm. The length of each tangent is 24 cm. Find the perimeter of the quadrilateral $PAOB$. 3

22. A sector of a circle of radius 21 cm has central angle 72°. Find the length of its arc. 3

23. If $\sin A = \dfrac{12}{13}$, find $\cos A$ and $\tan A$. 3

24. From the top of a cliff $15\sqrt{3}$ m high, the angle of depression of a boat on the sea is 60°. Find the horizontal distance of the boat from the foot of the cliff. 3

OR

Tangents PA and PB are drawn from an external point P to a circle of centre O and radius 6 cm. The length of each tangent is 8 cm. Find the area of the quadrilateral $PAOB$ (in cm^2). 3

25. A sector of a circle of radius 21 cm has central angle 90°. Find the perimeter of the sector. 3

OR

Evaluate: $\tan^2 45^\circ + \sin^2 90^\circ$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. If $\tan\theta = \dfrac{12}{5}$, find $\sin\theta$, $\cos\theta$ and $\sec\theta$. 5

OR

Two points on level ground lie on opposite sides of a tower of height $9\sqrt{3}$ m, in line with its foot. The angles of elevation of the top of the tower from the two points are 30° and 60°. Find the distance between the two points. 5

27. Tangents PA and PB are drawn from a point P at a distance of 10 cm from the centre O of a circle of radius 6 cm. Find the length of each tangent, the perimeter and the area of the quadrilateral $PAOB$ (in this order). 5

OR

A circular park of radius 21 m is surrounded by a path 7 m wide. Find the area of the path. 5

END OF QUESTION PAPER **Total: 50 Marks**

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